

then can take proper action on the failure event. It is possible that a more advanced PSE, which provides a management option of resetting the power connection, could revive the PD in time.

Diverse output voltage. Just like various control systems (such as Modbus, ProfitNet and DeviceNet) used in different industrial applications, the power voltage required by each industrial application may vary.

Take public transportation as an example. Here the power supply is usually 24V DC, which does not fall in the range defined in the IEEE802.3af standard (44V DC to 57V DC). An additional voltage transformer may solve this problem. However, it may not meet the requirements for some particular reasons, such as costs and space limitation. Being able to feed a wider range of power voltages from the PSE would be a more direct solution. This becomes another challenge for the PSE manufacturers to achieve the needs for diverse industrial applications.

Ultra-power demand and maximum power limitation. Driven by the benefits and the evolution of the technology, more and more PDs consuming high power are coming to the market. WiMax base stations and PZT cameras are such kinds of products already seen today. However, the amount of power drawn by the PD must not cause any damage to the PSE, the cable or the PD itself.

Power delivery generates heat and bundling of the cables may exacerbate the heating problem. For safety concern, if either the PSE or the PD is not well designed to comply with the standard, the power delivery should be well controlled by some means to avoid overheating.

DC disconnect. There may be PD products designed to enter sleeping mode in order to save power if they have been idle for a while. In the power-saving mode, the power consumption is less. However, if the power draw goes below the DC-

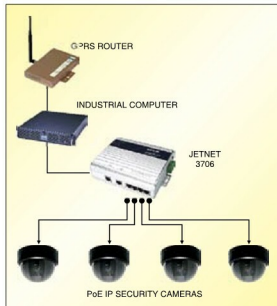


Fig. 4: PoE IP surveillance on bus

disconnect threshold, the PSE judges the PD removal, starts DC disconnect process and stops delivering power. The PD is then turned off and can never be woken up by its event triggers.

The problem may not be limited to power-saving mode. The PSE should provide some mechanisms to prevent unexpected DC disconnect in order to ensure the stability and reliability for industrial PoE applications.

Wide-ranged operating temperature. Industrial devices are often located outdoors or in control rooms, which could be extremely cold, hot or humid. So an industrial-grade PoE device should be able to operate under harsh conditions.

In particular, feeding power generates heat. The more the power is delivered, the higher the operating temperature of the PoE device. With the ambient heat in the environment, the high-temperature circumstances may get from bad to worse. High-temperature tolerance and over-heating protection are two important requirements for industrial PoE.

Advanced monitoring and management. In industrial environments, it may not be enough to power-on a

PoE device and let it run by itself. Advanced monitoring and management of each PD, PSE or even port of a PSE is important to the IT staff for many system maintenance requirements.

For example, polling the amount of real-time power delivery on each port of a PSE helps to monitor the current status and get an idea of the trend of the PoE port.

Abnormal conditions such as power cut, over-feeding or under-supply can then be observed in advance to prevent errors.

Error detection can also

be achieved by event-trigger approach. By setting a certain circumstance of threshold, any irregular event exceeding the circumstance will trigger a notification to the IT staff or result in a predefined action. Notifications can be in the form of alarm, e-mail, SNMP trap or other means.

Accompanied with notifications, a predefined action can take place immediately for protection or recovery purposes before the IT staff receives the notification and arrives for on-site service. Furthermore, detailed power control over a PoE system can also be used for security or power saving.

Scheduling power control is one of the methods to save power and utilise the resource economically. Advanced monitoring and management—which enhance the reliability, security and efficiency of a PoE system—are essential for industrial environments.

Real-world industrial PoE applications

Along with the conveniences and advantages, more and more vendors recognise the applications which the PoE technology can be applied to. Here we cite examples of how PoE solutions